



“EMPOWERMENT THROUGH TECHNOLOGICAL EXCELLENCE”

GENBA SOPANRAO MOZE COLLEGE OF ENGINEERING

S No 25/1/3, Balewadi-Baner, Pune – 411 045

(Approved by AICTE and Govt. of Maharashtra, Affiliated to Savitribai Phule Pune University)

DTE Code - EN6144 University Affiliation ID - PU/PN/ENGG/138/1999

Ph: 020-27390500

Website: www.gsmozecoe.co.in

Email: gsmoze@yahoo.co.in

Founder President: Shri Rambhau Moze

Department of Information Technology

LAB MANUAL OF

Digital Marketing and social media

Course Code: VSE- 281-ITT

COURSE: I.T. Engineering

Class: SE 2024 PAT

Title

1. Keyword Research and SEO Optimization using SEMrush

Aim

To perform keyword research for a specific industry/topic using SEO tools (SEMrush) and to propose effective on-page and off-page SEO optimization strategies for a hypothetical website based on keyword research findings.

Learning Objectives

After completing this lab, students will be able to: - Understand the importance of keyword research in SEO - Use SEMrush for identifying high-volume and relevant keywords - Analyze keyword metrics such as search volume, keyword difficulty, CPC, and intent - Design on-page SEO strategies based on selected keywords - Propose off-page SEO techniques to improve website authority and rankings

Industry / Topic Selected

Industry: Renewable Energy Solutions **Hypothetical Website** GreenFutureEnergy.com

Website Goal: Generate leads and awareness for solar and wind energy solutions in India

Tools Required

- SEMrush (Free or Trial version)
 - Web Browser
 - Google Search Engine
 - Spreadsheet software (Excel / Google Sheets)
-

Theory

What is Keyword Research?

Keyword research is the process of identifying words and phrases that users type into search engines while looking for information, products, or services. These keywords help businesses optimize their content to match user intent and improve visibility on search engine result pages (SERPs).

Importance of SEO

Search Engine Optimization (SEO) improves a website's organic visibility, drives targeted traffic, enhances brand credibility, and increases conversions without paid advertising.

SEO is broadly classified into: - **On-page SEO** – Optimization within the website - **Off-page SEO** – External optimization techniques

Procedure

Step 1: Access SEMrush

1. Open www.semrush.com
2. Login or sign up for a free account
3. Navigate to **Keyword Overview** or **Keyword Magic Tool**

Step 2: Seed Keyword Identification

Enter seed keywords related to the selected industry: - renewable energy - solar energy solutions - wind energy in India

Step 3: Keyword Analysis

Analyze keywords based on: - Search Volume - Keyword Difficulty (KD) - Cost Per Click (CPC) - Search Intent (Informational, Commercial, Transactional)

Step 4: Keyword Selection

Select high-volume, low-to-medium difficulty keywords that are most relevant to the website goals.

Keyword Research Findings

High-Volume & Relevant Keywords

Keyword	Search Volume	Keyword Difficulty	Intent
renewable energy	90,000	High	Informational
solar energy solutions	40,000	Medium	Commercial
solar panel installation	33,000	Medium	Transactional
wind energy in India	18,000	Medium	Informational
green energy companies	22,000	Medium	Commercial
rooftop solar system	27,000	Medium	Transactional
renewable energy benefits	14,000	Low	Informational
solar power for homes	19,000	Medium	Commercial

On-Page SEO Optimization Strategy

1. Title Tag Optimization

- Include primary keyword at the beginning
- Example: **Solar Energy Solutions in India | GreenFuture Energy**

2. Meta Description

- Use keywords naturally
- Keep under 160 characters

- Example: *Affordable solar energy solutions for homes and businesses in India. Get clean and renewable energy today.*

3. URL Structure

- SEO-friendly URLs
- Example: www.greenfutureenergy.com/solar-energy-solutions

4. Header Tags (H1–H3)

- H1: Solar Energy Solutions for Sustainable Future
- H2: Benefits of Renewable Energy
- H3: Rooftop Solar System for Homes

5. Content Optimization

- High-quality, informative content
- Keyword density: 1–2%
- Use LSI keywords

6. Image Optimization

- Use descriptive file names
- Add ALT text with keywords
- Compress images for faster loading

7. Internal Linking

- Link related blog posts and service pages
- Improves crawlability and user experience

8. Mobile & Page Speed Optimization

- Mobile-responsive design
- Improve Core Web Vitals

Off-Page SEO Optimization Strategy

1. Backlink Building

- Guest blogging on renewable energy websites
- Industry directories
- Educational (.edu) backlinks

2. Social Media Marketing

- Share blogs on LinkedIn, Twitter, and Facebook
- Promote renewable energy awareness campaigns

3. Local SEO

- Google Business Profile optimization
- Local citations (Justdial, IndiaMart)

4. Influencer Outreach

- Collaborate with sustainability influencers
- Product reviews and mentions

5. Content Marketing

- Publish blogs, case studies, and white papers
 - Infographics related to green energy
-

Result

High-volume and relevant keywords were successfully identified using SEMrush. Based on the findings, comprehensive on-page and off-page SEO optimization strategies were proposed for the hypothetical website GreenFutureEnergy.com.

Conclusion

Keyword research is the backbone of effective SEO. Using tools like SEMrush enables data-driven decision-making for content creation and optimization. Proper implementation of on-page and off-page SEO strategies can significantly improve website visibility, organic traffic, and lead generation.

Viva Voce Questions

1. What is keyword research?
 2. Explain keyword difficulty.
 3. Difference between on-page and off-page SEO.
 4. What is search intent?
 5. Name any two SEO tools.
-

References

1. SEMrush Official Documentation
2. Google SEO Starter Guide
3. Moz Beginner's Guide to SEO

Title

2. Design and Development of a Social Media Content Calendar

Aim

To design a structured social media content calendar for a selected organization/brand in order to plan, organize, and schedule social media posts effectively across multiple platforms.

Learning Objectives

After completing this lab, students will be able to: - Understand the concept and importance of a social media content calendar - Identify content goals and target audience - Plan platform-wise social media content - Align content with marketing objectives and brand voice - Improve consistency, engagement, and reach on social media platforms

Industry / Topic Selected

Industry: Educational Technology (EdTech)

Hypothetical Brand: SmartLearn Academy

Platforms Used: Instagram, Facebook, LinkedIn, Twitter (X)

Business Objective: - Increase brand awareness - Drive website traffic - Generate student enrollments

Tools Required

- Microsoft Excel / Google Sheets
 - Canva (for creatives)
 - Meta Business Suite (optional)
 - Buffer / Hootsuite (optional scheduling tools)
 - Internet Browser
-

Theory

What is a Social Media Content Calendar?

A social media content calendar is a planning document that outlines what content will be posted, on which platform, and on what date and time. It helps marketers maintain consistency, plan campaigns in advance, and track performance.

Importance of a Content Calendar

- Ensures consistent posting
- Saves time and resources

- Aligns content with marketing goals
- Helps analyze content performance
- Avoids last-minute content creation

Types of Social Media Content

- Educational posts
 - Promotional posts
 - Engagement posts (polls, quizzes)
 - Informational posts
 - Testimonials & success stories
 - Festival & event-based posts
-

Procedure

Step 1: Define Goals

Identify clear objectives such as brand awareness, lead generation, or engagement.

Step 2: Identify Target Audience

- Students (18–25 years)
- Working professionals
- Trainers and educators

Step 3: Select Platforms

Choose platforms based on audience behavior: - Instagram – Visual & reels - Facebook – Community engagement - LinkedIn – Professional & academic updates - Twitter (X) – Announcements & trends

Step 4: Decide Content Themes

- Motivation Monday
- Tech Tip Tuesday
- Webinar Wednesday
- Feature Friday
- Success Story Sunday

Step 5: Create Content Calendar

Prepare a weekly or monthly schedule with post details.

Sample Weekly Social Media Content Calendar

Day	Platform	Content Type	Description	Goal
-----	----------	--------------	-------------	------

Day	Platform	Content Type	Description	Goal
Monday	Instagram	Motivational Post	Learning motivation quote	Engagement
Tuesday	LinkedIn	Educational	AI & EdTech trends	Awareness
Wednesday	Instagram	Reel	Online course demo	Promotion
Thursday	Twitter	Poll	Preferred learning method	Engagement
Friday	Facebook	Testimonial	Student success story	Trust
Saturday	Instagram	Carousel	Course benefits	Lead Generation
Sunday	Facebook	Announcement	Upcoming webinar	Registration

Content Guidelines

1. Visual Consistency

- Use brand colors and fonts
- Maintain logo placement

2. Caption Writing

- Clear and concise
- Use CTA (Call to Action)
- Relevant hashtags

3. Hashtag Strategy

- Branded hashtags (#SmartLearn)
- Industry hashtags (#EdTech, #OnlineLearning)
- Trending hashtags

4. Posting Frequency

- Instagram: 4–5 posts/week
 - Facebook: 3–4 posts/week
 - LinkedIn: 2–3 posts/week
 - Twitter: Daily updates
-

Performance Metrics

- Likes, shares, comments
- Reach and impressions
- Click-through rate (CTR)

- Follower growth
-

Result

A structured social media content calendar was successfully created for the hypothetical EdTech brand SmartLearn Academy, enabling systematic planning and effective content distribution.

Conclusion

A social media content calendar plays a vital role in digital marketing by ensuring consistency, clarity, and strategic alignment. Proper planning helps brands achieve better engagement, audience retention, and marketing success.

Applications

- Digital marketing campaigns
 - Brand promotion
 - Educational institutions
 - Startups and small businesses
-

Viva Voce Questions

1. What is a social media content calendar?
 2. Why is content planning important?
 3. Name any two social media scheduling tools.
 4. What are engagement metrics?
 5. What is CTA?
-

References

1. Meta Business Help Center
2. HubSpot Social Media Marketing Guide
3. Hootsuite Blog

Title

3. Design and Execution of an Email Marketing Campaign

Aim

To design, plan, and analyze an email marketing campaign for a selected organization using appropriate email marketing strategies and tools.

Learning Objectives

After completing this lab, students will be able to: - Understand the fundamentals of email marketing - Identify target audience and campaign goals - Design effective email content and structure - Apply best practices for subject lines and CTAs - Analyze email campaign performance metrics

Industry / Topic Selected

Industry: E-commerce

Hypothetical Brand: ShopEase Online Store

Campaign Objective: Promote seasonal discounts and increase online sales

Target Audience: - Online shoppers (18–45 years) - Price-sensitive customers - Existing and new subscribers

Tools Required

- Email Marketing Tool (Mailchimp / Sendinblue / HubSpot – demo)
 - Microsoft Word / Google Docs
 - Spreadsheet (for email list management)
 - Internet Browser
-

Theory

What is Email Marketing?

Email marketing is a digital marketing technique that involves sending targeted emails to a group of subscribers to promote products, services, or build customer relationships.

Importance of Email Marketing

- Cost-effective marketing channel
- Personalized communication

- High ROI compared to other digital channels
 - Helps in customer retention and engagement
-

Types of Email Campaigns

- Promotional Emails
 - Newsletter Emails
 - Transactional Emails
 - Welcome Emails
 - Re-engagement Emails
-

Procedure

Step 1: Define Campaign Goal

Determine the objective such as sales promotion, lead nurturing, or brand awareness.

Step 2: Create or Segment Email List

- Existing customers
- New subscribers
- Inactive users

Step 3: Design Email Content

Include: - Subject line - Header image - Email body content - Call-to-Action (CTA) - Footer with contact details

Step 4: Personalization

Use customer name, location, or preferences to personalize emails.

Step 5: Schedule and Send Email

Choose optimal date and time for sending emails.

Step 6: Monitor Campaign Performance

Track key performance indicators.

Sample Email Campaign Structure

Subject Line Examples

- “Flat 30% Off on Your Favorite Products!”
- “Exclusive Deals Just for You – ShopEase”

Email Body (Sample)

- Greeting with recipient name
 - Highlight discount offers
 - Product images and short descriptions
 - Strong CTA button (“Shop Now”)
-

Best Practices in Email Marketing

- Use clear and compelling subject lines
 - Avoid spam trigger words
 - Maintain proper email frequency
 - Ensure mobile responsiveness
 - Provide unsubscribe option
-

Performance Metrics

Metric	Description
Open Rate	Percentage of emails opened
Click Through Rate (CTR)	Percentage of clicks on links
Conversion Rate	Purchases completed
Bounce Rate	Failed email deliveries
Unsubscribe Rate	Users opting out

Result

An effective email marketing campaign was designed for the hypothetical e-commerce brand ShopEase, covering campaign planning, content creation, and performance evaluation.

Conclusion

Email marketing remains one of the most powerful digital marketing tools. A well-planned and targeted email campaign helps businesses increase conversions, retain customers, and build long-term relationships.

Applications

- E-commerce promotions
- Educational institution announcements

- Startup marketing campaigns
 - Customer engagement programs
-

Viva Voce Questions

1. What is email marketing?
 2. What is CTR?
 3. Difference between promotional and transactional emails.
 4. Name any two email marketing tools.
 5. What is email personalization?
-

References

1. Mailchimp Email Marketing Guide
2. HubSpot Email Marketing Resources
3. Digital Marketing Institute

Title

4. Design and Execution of a Social Media Advertising Campaign using Facebook Business Manager

Aim

To create and manage a social media advertising campaign using Facebook Business Manager by defining campaign objectives, target audience, ad formats, budget allocation, and monitoring campaign performance metrics.

Learning Objectives

After completing this lab, students will be able to: - Understand Facebook Business Manager and its features - Create and configure a social media advertising campaign - Define campaign objectives and audience targeting criteria - Select suitable ad formats and budget strategies - Monitor and analyze campaign performance metrics

Industry / Topic Selected

Industry: Educational Services (EdTech)

Hypothetical Brand: SmartLearn Academy

Advertising Platforms: Facebook & Instagram

Campaign Goal: Increase course enrollments

Tools Required

- Facebook Business Manager account
 - Facebook Ads Manager
 - Internet Browser
 - Canva (for ad creatives – optional)
 - Spreadsheet (for performance tracking)
-

Theory

What is Social Media Advertising?

Social media advertising refers to paid promotional activities on social platforms such as Facebook and Instagram to reach a specific target audience. These ads are highly customizable based on demographics, interests, behavior, and location.

Facebook Business Manager

Facebook Business Manager is a centralized platform that allows businesses to manage pages, ad accounts, campaigns, audiences, and performance analytics in one place.

Campaign Objectives

The campaign objectives define what action the advertiser wants users to take.

Selected Objective: Lead Generation / Conversions

Other objectives include: - Brand Awareness - Reach - Traffic - Engagement - App Installs

Procedure

Step 1: Create Facebook Business Manager Account

1. Visit business.facebook.com
2. Click on “Create Account”
3. Enter business name, email ID, and details
4. Add Facebook Page and Ad Account

Step 2: Create a New Ad Campaign

1. Open Facebook Ads Manager
2. Click on “Create Campaign”
3. Select campaign objective (Conversions / Leads)
4. Name the campaign

Step 3: Define Target Audience

Audience Targeting Criteria: - Location: India - Age Group: 18–30 years - Gender: All - Interests: Online learning, Technology, Engineering, Competitive exams - Education Level: College students & graduates

Ad Formats Used

Ad Format	Description
Image Ads	Single image with CTA
Video Ads	Short promotional videos
Carousel Ads	Multiple course highlights
Story Ads	Full-screen vertical ads

Budget Allocation

Campaign Duration: 7 days

Level	Budget
Daily Budget	₹500 per day
Total Budget	₹3,500

Bidding Strategy: Lowest Cost

Ad Creative Description

- Headline: “Upgrade Your Skills with SmartLearn Academy”
- Primary Text: “Join industry-ready courses and boost your career today.”
- CTA Button: “Enroll Now”
- Landing Page: Course registration page

Campaign Performance Metrics

Metric	Description
Reach	Number of unique users who saw the ad
Impressions	Total number of times the ad was shown
Engagement	Likes, comments, shares
CTR	Click-through rate
Conversion Rate	Successful enrollments
CPC	Cost per click

Sample Performance Analysis (Hypothetical)

Metric	Value
Reach	18,000 users
Engagement	1,450 interactions
CTR	3.2%
Conversions	95 enrollments
CPC	₹12

Result

A successful social media advertising campaign was created and managed using Facebook Business Manager. Campaign objectives, audience targeting, ad formats, budget allocation, and performance metrics were effectively defined and analyzed.

Conclusion

Social media advertising using Facebook and Instagram enables precise audience targeting and measurable results. Proper planning, creative design, and performance monitoring help businesses achieve higher engagement and conversions.

Applications

- Course promotions
 - Product launches
 - Event registrations
 - Brand awareness campaigns
-

Viva Voce Questions

1. What is Facebook Business Manager?
 2. Explain campaign objectives in Facebook Ads.
 3. What is CTR?
 4. Name any two Facebook ad formats.
 5. What is audience targeting?
-

References

1. Facebook Business Help Center
2. Meta Ads Manager Documentation
3. HubSpot Social Media Advertising Guide

Title

5. Leveraging Instagram Influencers for Brand Promotion

Aim

To design an Instagram influencer outreach plan for promoting a new line of eco-friendly skincare products with the objective of increasing brand visibility and driving online sales among environmentally conscious consumers.

Learning Objectives

After completing this lab, students will be able to: - Understand influencer marketing concepts - Identify suitable Instagram influencers based on brand goals - Design an influencer outreach and collaboration plan - Define campaign objectives, content strategy, and budget - Measure influencer campaign performance metrics

Industry / Product Selected

Industry: Beauty & Personal Care

Product Line: EcoGlow – Eco-friendly Skincare Products

Platform: Instagram

Target Market: Environmentally conscious consumers

Brand Overview

EcoGlow is a sustainable skincare brand offering products made from natural ingredients, cruelty-free testing, recyclable packaging, and eco-friendly manufacturing processes.

Tools Required

- Instagram (Business Account)
 - Influencer research tools (Heepsy / HypeAuditor – demo)
 - Google Sheets / Excel
 - Canva (for creatives)
 - Internet Browser
-

Theory

What is Influencer Marketing?

Influencer marketing is a digital marketing strategy that involves collaborating with individuals who have a strong presence and influence on social media platforms to promote products or services.

Importance of Instagram Influencers

- High engagement rates
 - Visual storytelling
 - Strong impact on purchase decisions
 - Authentic brand promotion
-

Campaign Objectives

- Increase brand awareness and reach
 - Build trust and credibility
 - Drive traffic to the website
 - Increase online sales
-

Target Audience Profile

Criteria	Description
Age	18–35 years
Gender	All
Location	Urban India
Interests	Skincare, sustainability, vegan lifestyle
Behavior	Online shoppers, eco-conscious

Influencer Selection Criteria

Parameter	Requirement
Influencer Type	Micro & Nano Influencers
Followers	10K – 100K
Engagement Rate	>3%
Niche	Skincare / Sustainability
Authenticity	Real followers & content

Influencer Outreach Plan

Step 1: Identify Influencers

Search hashtags like #EcoSkincare, #CleanBeauty, #SustainableLiving

Step 2: Shortlist Influencers

Evaluate based on engagement, content quality, audience relevance

Step 3: Outreach Message (Sample)

“Hi [Name], we love your content on sustainable skincare. We’d love to collaborate with you to promote our eco-friendly skincare range EcoGlow. Let us know if you’re interested!”

Step 4: Collaboration Type

- Product reviews
- Reels & stories
- Giveaway campaigns
- Affiliate links

Content Strategy

Content Type	Description
Reels	Product application & benefits
Stories	Daily skincare routine
Posts	Before/after results
Live Sessions	Q&A on eco skincare

Budget Allocation (Hypothetical)

Influencer Type	Cost per Influencer
Nano Influencer	₹3,000
Micro Influencer	₹8,000

Total Budget: ₹50,000

Performance Metrics

Metric	Description
Reach	Number of users exposed
Engagement	Likes, comments, shares
Website Clicks	Traffic generated
Conversion Rate	Sales completed

Metric	Description
ROI	Revenue vs cost

Sample Campaign Outcome (Hypothetical)

Metric	Result
Total Reach	1,20,000
Engagement Rate	4.5%
Website Visits	3,000
Sales	220 orders

Result

An Instagram influencer outreach plan was successfully designed for promoting eco-friendly skincare products. The plan focused on influencer selection, content strategy, budget allocation, and performance measurement.

Conclusion

Influencer marketing on Instagram is an effective strategy for promoting eco-friendly brands. Collaborating with niche influencers helps build trust, increase brand visibility, and drive meaningful conversions.

Applications

- Sustainable product promotion
- Brand awareness campaigns
- E-commerce sales growth
- Community building

Viva Voce Questions

1. What is influencer marketing?
2. Difference between nano and micro influencers.
3. Why is Instagram effective for influencer campaigns?
4. What is engagement rate?
5. How do influencers drive brand trust?

References

1. Influencer Marketing Hub

2. Instagram Business Help Center
3. HubSpot Influencer Marketing Guide

Title

6. Creating Canva Designs for YouTube Channel Branding

Aim

To design professional branding graphics for a YouTube cooking tutorial channel named “**Tasty Bites Kitchen**” using Canva, including a channel banner, profile picture (logo), and a reusable thumbnail template.

Learning Objectives

After completing this lab, students will be able to: - Understand the importance of branding for YouTube channels - Use Canva for creating digital graphics - Design a YouTube channel banner, logo, and thumbnail template - Apply design principles such as color, typography, and layout - Create reusable templates for consistent branding

Channel Overview

Channel Name: Tasty Bites Kitchen

Channel Category: Cooking Tutorials

Target Audience: Home cooks, food enthusiasts, beginners (18–45 years)

Brand Personality: Warm, tasty, friendly, and visually appealing

Tools Required

- Canva (Free or Pro version)
 - Internet Browser
 - YouTube (for size reference)
 - Laptop / Desktop System
-

Theory

Importance of YouTube Channel Branding

YouTube channel branding helps create a strong first impression, improves recognition, and builds trust among viewers. Consistent visuals increase subscriber engagement and channel recall.

Canva as a Design Tool

Canva is an easy-to-use online graphic design platform that provides pre-designed templates, fonts, icons, and images suitable for social media and video branding.

Design Specifications

1. YouTube Channel Banner

- Size: 2560 × 1440 pixels
- Safe Area: 1546 × 423 pixels (center)
- Purpose: Communicate channel theme and upload schedule

2. Profile Picture (Channel Logo)

- Size: 800 × 800 pixels
- Shape: Circular (displayed)
- Purpose: Brand identity

3. Video Thumbnail Template

- Size: 1280 × 720 pixels
 - Aspect Ratio: 16:9
 - Purpose: Attract clicks and maintain consistency
-

Procedure

Step 1: Login to Canva

1. Open www.canva.com
2. Login or sign up using email or Google account

Step 2: Create Channel Banner

3. Search for “YouTube Channel Art” in Canva
4. Select a suitable cooking-themed template
5. Customize with:
 - Channel name: *Tasty Bites Kitchen*
 - Tagline: *Simple & Delicious Recipes*
 - Food images and icons
6. Apply warm colors (red, orange, yellow)
7. Download in PNG format

Step 3: Create Profile Picture (Logo)

8. Search “Logo” or “YouTube Profile Picture”
9. Use food icons (chef hat, spoon, plate)
10. Add initials or full channel name
11. Use minimal text and bold colors
12. Download as PNG

Step 4: Create Thumbnail Template

13. Search “YouTube Thumbnail”
14. Design a reusable layout with:
 - Large food image
 - Bold recipe title text

- Brand color overlay

15. Save as a template for future videos

Design Guidelines

- Use consistent color palette
 - Choose readable fonts
 - Avoid cluttered layouts
 - Highlight food images clearly
 - Maintain brand consistency
-

Sample Design Description

Graphic	Description
Banner	Cooking images with channel name & tagline
Logo	Circular logo with food icon & text
Thumbnail	Bold text, food image, brand colors

Result

Professional YouTube branding graphics including a channel banner, profile picture, and thumbnail template were successfully designed for the cooking channel “Tasty Bites Kitchen” using Canva.

Conclusion

Canva provides an efficient platform for creating attractive YouTube branding graphics without advanced design skills. Consistent branding helps improve channel recognition and viewer engagement.

Applications

- YouTube content creation
 - Personal branding
 - Food vlog channels
 - Digital marketing projects
-

Viva Voce Questions

16. What is YouTube channel branding?
17. Why are thumbnails important?
18. What is the safe area in YouTube banners?

19. Name any two features of Canva.

20. Why is consistency important in design?

References

21. Canva Design School

22. YouTube Creator Academy

23. HubSpot Visual Branding Guide

Title

7. Understanding Web Analytics using Google Analytics

Aim

To analyze website performance using Google Analytics and identify key areas for optimization and improvement based on user behavior and traffic data.

Learning Objectives

After completing this lab, students will be able to: - Understand the concept of web analytics - Use Google Analytics to track website performance - Analyze user behavior, traffic sources, and engagement metrics - Interpret key reports and dashboards - Identify optimization opportunities to improve website performance

Website / Industry Selected

Industry: Educational Website (EdTech)

Hypothetical Website: www.smartlearnacademy.com

Website Objective: - Increase website traffic - Improve user engagement - Generate course inquiries and enrollments

Tools Required

- Google Analytics (GA4)
 - Google Account
 - Internet Browser
 - Spreadsheet software (Excel / Google Sheets)
-

Theory

What is Web Analytics?

Web analytics is the process of collecting, measuring, analyzing, and reporting website data to understand user behavior and optimize website performance.

Google Analytics

Google Analytics is a powerful free web analytics tool provided by Google that helps track website traffic, user interactions, conversions, and overall performance.

Key Web Analytics Metrics

Metric	Description
Users	Number of unique visitors
Sessions	Total visits to the website
Page Views	Total pages viewed
Bounce Rate	Percentage of users leaving after one page
Engagement Rate	User interaction level
Conversion Rate	Completed goals

Procedure

Step 1: Create Google Analytics Account

1. Visit analytics.google.com
2. Sign in using Google account
3. Click on “Start Measuring”
4. Create a new GA4 property
5. Add website URL and business details

Step 2: Install Tracking Code

6. Copy the GA4 Measurement ID
7. Add tracking code to website header
8. Verify data collection status

Step 3: Access Google Analytics Dashboard

- Navigate to Home Dashboard
- Monitor real-time user activity

Step 4: Analyze Audience / User Reports

Metrics Analyzed: - Total users - New vs returning users - Geographic location - Device type (mobile, desktop)

Step 5: Analyze Traffic Acquisition

Source	Description
Organic Search	Search engine traffic
Direct	Direct URL visits
Referral	External website links
Social	Social media traffic

Source	Description
Paid Search	Advertisements

Step 6: Analyze Engagement Reports

- Average engagement time
 - Pages per session
 - Most visited pages
 - Exit pages
-

Step 7: Analyze Conversion Events

- Form submissions
 - Course inquiry clicks
 - Registration completions
-

Sample Performance Analysis (Hypothetical)

Metric	Value
Users	12,500
Sessions	18,200
Bounce Rate	48%
Engagement Rate	62%
Conversion Rate	3.8%

Areas Identified for Optimization

1. High Bounce Rate Pages

- Improve content quality
- Add internal links
- Optimize page load speed

2. Low Conversion Pages

- Improve CTA placement
- Simplify forms
- Add trust signals

3. Mobile Optimization

- Improve mobile responsiveness
- Reduce image size

4. Traffic Source Improvement

- Increase SEO efforts

- Optimize social media campaigns
-

Recommendations for Improvement

- Enhance content relevance
 - Improve website speed
 - Optimize landing pages
 - Implement A/B testing
 - Track custom events
-

Result

Website performance was successfully analyzed using Google Analytics. Key user behavior metrics were studied and several optimization opportunities were identified to improve engagement and conversions.

Conclusion

Google Analytics provides valuable insights into website performance and user behavior. Regular analysis helps organizations make data-driven decisions to enhance user experience and achieve business goals.

Applications

- Website performance monitoring
 - Digital marketing optimization
 - UX improvement
 - Conversion rate optimization
-

Viva Voce Questions

9. What is web analytics?
 10. What is bounce rate?
 11. Difference between users and sessions.
 12. Name any two Google Analytics reports.
 13. What is conversion tracking?
-

References

14. Google Analytics Help Center
15. Google Skillshop – Analytics
16. Moz Web Analytics Guide

Title

8. Creating a Website using WordPress and Plugins

Aim

To create a functional website using WordPress, install and configure different plugins including social media plugins and page builder plugins (Elementor), learn how to insert sections using Elementor, and understand the creation of a website using Google Sites.

Learning Objectives

After completing this lab, students will be able to: - Understand WordPress as a Content Management System (CMS) - Create a website using WordPress - Install and use different WordPress plugins - Use social media plugins for integration - Design web pages using Elementor page builder - Create basic websites using Google Sites

Tools / Software Required

- Internet Browser (Chrome / Firefox)
 - WordPress (wordpress.com or localhost)
 - Elementor Plugin
 - Social Media Plugins (Jetpack / Social Icons)
 - Google Account (for Google Sites)
-

Theory

What is WordPress?

WordPress is an open-source Content Management System (CMS) used to create websites and blogs easily without extensive coding knowledge. It supports themes and plugins to extend website functionality.

What are Plugins?

Plugins are software components that add specific features and functionality to a WordPress website such as SEO, security, social media sharing, page building, and performance optimization.

Types of WordPress Plugins

Plugin Type	Purpose
-------------	---------

Plugin Type	Purpose
SEO Plugins	Improve search engine visibility
Security Plugins	Protect website from threats
Social Media Plugins	Share & integrate social platforms
Page Builder Plugins	Drag-and-drop page design

Procedure – Part A: Creating Website using WordPress

Step 1: Create WordPress Website

1. Visit www.wordpress.com
 2. Click on “Get Started”
 3. Create account using email
 4. Choose website name and category
 5. Select a free theme
 6. Launch the website
-

Procedure – Part B: Installing WordPress Plugins

Step 2: Install Plugins

1. Login to WordPress dashboard
 2. Go to Plugins → Add New
 3. Search required plugin
 4. Click Install → Activate
-

Commonly Used Plugins

Plugin Name	Function
Elementor	Page builder
Jetpack	Security & performance
Yoast SEO	Search optimization
Social Media Share Buttons	Social sharing

Procedure – Part C: Social Media Plugins

Step 3: Install Social Media Plugin

1. Go to Plugins → Add New
2. Search “Social Media Icons”
3. Install and activate plugin
4. Add social media links (Facebook, Instagram, Twitter)
5. Display icons on header/footer

Procedure – Part D: Page Builder Plugin – Elementor

What is Elementor?

Elementor is a drag-and-drop page builder plugin that allows users to design web pages visually without coding.

Step 4: Install Elementor

1. Go to Plugins → Add New
 2. Search “Elementor”
 3. Click Install → Activate
-

Step 5: Create Page using Elementor

1. Go to Pages → Add New
 2. Click “Edit with Elementor”
 3. Elementor editor opens
-

Step 6: Insert a Section in Elementor

1. Click on “+” icon
 2. Select column structure (1, 2, or 3 columns)
 3. Drag widgets (Text, Image, Button)
 4. Customize content, color, and alignment
 5. Click Publish
-

Elementor Widgets Used

- Heading
 - Text Editor
 - Image
 - Button
 - Video
 - Icon List
-

Procedure – Part E: Creating Website using Google Sites

What is Google Sites?

Google Sites is a free website creation tool by Google that allows users to build simple websites without coding.

Step 7: Create Website using Google Sites

1. Visit sites.google.com
 2. Click on “+ Blank” site
 3. Enter site name and page title
 4. Add text, images, and layouts
 5. Create multiple pages
 6. Click Publish
-

Comparison: WordPress vs Google Sites

Feature	WordPress	Google Sites
Customization	High	Limited
Plugins	Yes	No
Ease of Use	Moderate	Very Easy
Best For	Professional websites	Basic websites

Result

A website was successfully created using WordPress with the help of plugins such as social media plugins and Elementor page builder. Additionally, a basic website was created using Google Sites.

Conclusion

WordPress is a powerful CMS that allows extensive customization through plugins and page builders like Elementor. Google Sites provides a simple alternative for quick website creation. Both platforms are useful depending on user requirements.

Applications

- Business websites
 - Personal blogs
 - College websites
 - Portfolio sites
-

Viva Voce Questions

1. What is WordPress?
2. What are plugins?

3. What is Elementor?
 4. Difference between WordPress and Google Sites.
 5. Name any two social media plugins.
-

References

1. WordPress Official Documentation
2. Elementor Help Center
3. Google Sites Help